

# evaluate\_binary\_log\_loss.R

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| evaluate_binary_log_loss        |
| <i>Evaluate Binary Log Loss</i> |

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## Description

Calculates mean binary cross-entropy after clipping probabilities away from zero and one.

## Usage

```
evaluate_binary_log_loss(  
  observed = NULL,  
  predicted_probability = NULL,  
  epsilon = 1e-06  
)
```

## Arguments

|                       |                                                                                        |
|-----------------------|----------------------------------------------------------------------------------------|
| observed              | Numeric binary observation vector.                                                     |
| predicted_probability | Numeric predicted-probability vector aligned with 'observed'.                          |
| epsilon               | Single numeric clipping tolerance strictly between zero and '0.5'. Defaults to '1e-6'. |

## Value

One-row tibble containing 'log\_loss', 'metric\_status', observation counts, prevalence, and the clipping tolerance. Valid one-class inputs retain a finite log loss because this metric does not require both classes.

**Examples**

```
evaluate_binary_log_loss(  
  observed = c(0, 1),  
  predicted_probability = c(0.25, 0.75)  
)
```

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